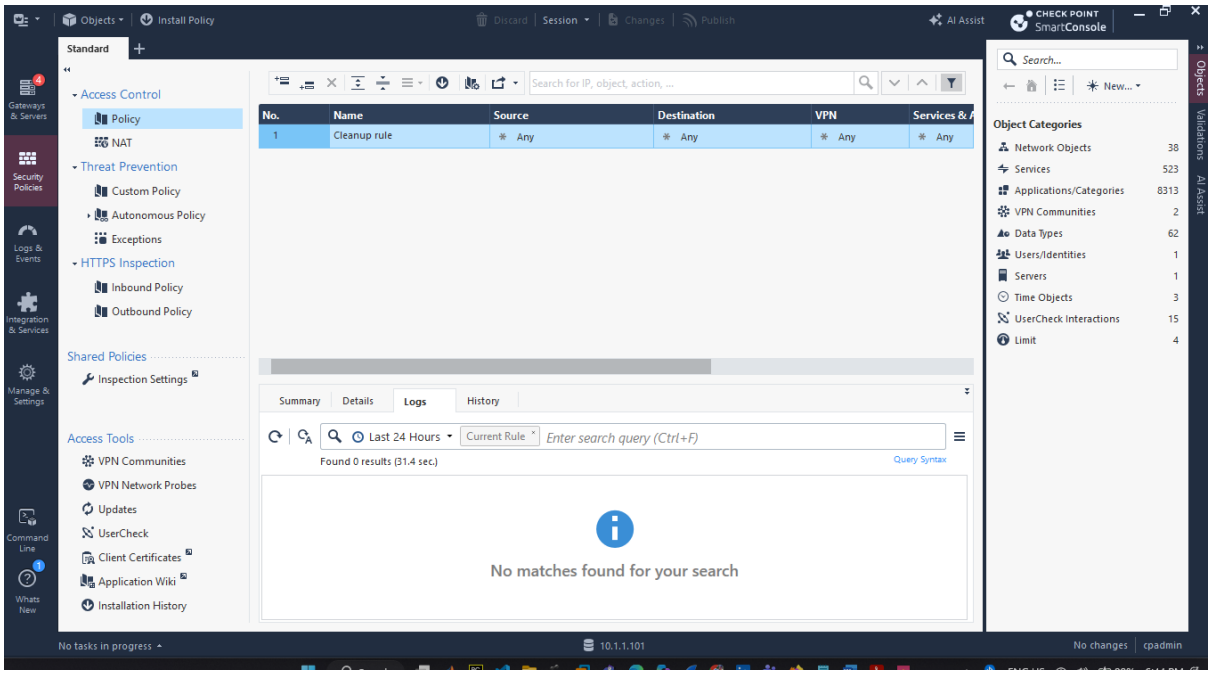
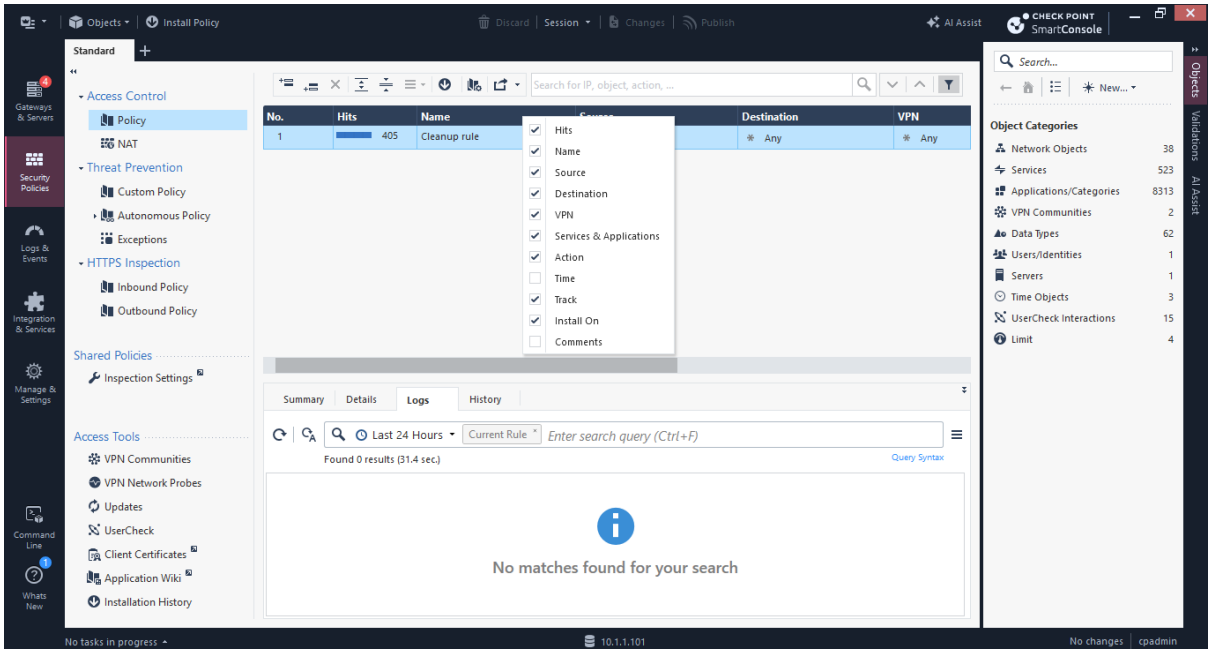
FULL NAME	**MUSA SHUAIBU**
REGISTRATION NUMBER	**C11/26/CCSA/17045**
SMARTCONSOLE AND A-SMS SERVER	**ICDFA provided**
VM HOSTNAME	**A-SMS and SmartConsule**

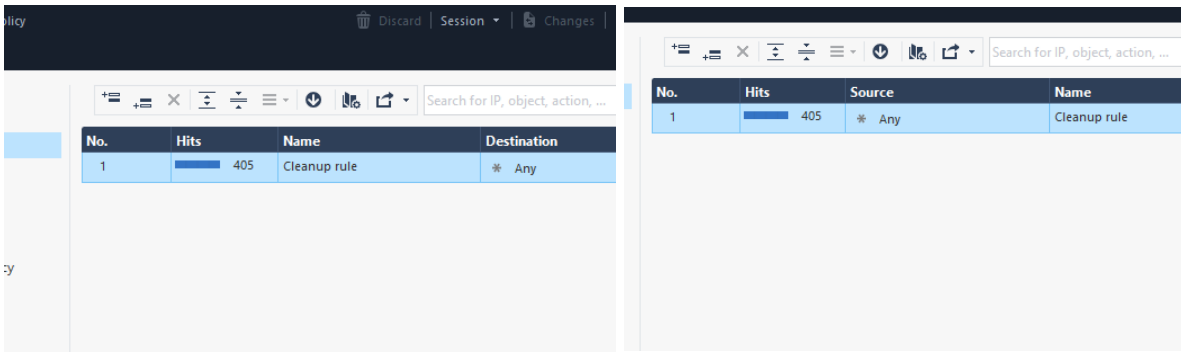
TASK 1: EXPLORE THE DEFAULT POLICY



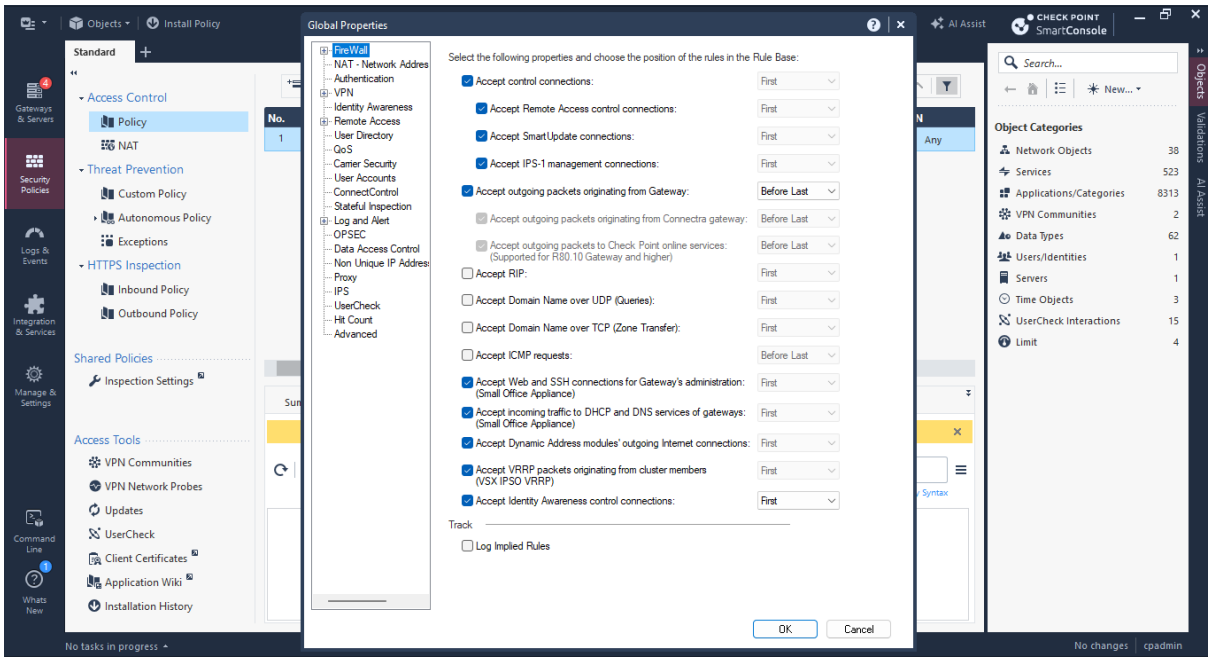
Brief Explanation: The Police package in use is a **cleanup rule** which can be configured to allow or drop new connections. It contains the following configuration
Name = cleanup rule, Source = Any, Destination = Any, VPN = Any, Service Application = Any, Action = Accept, Track = logs, Install on = Policy target



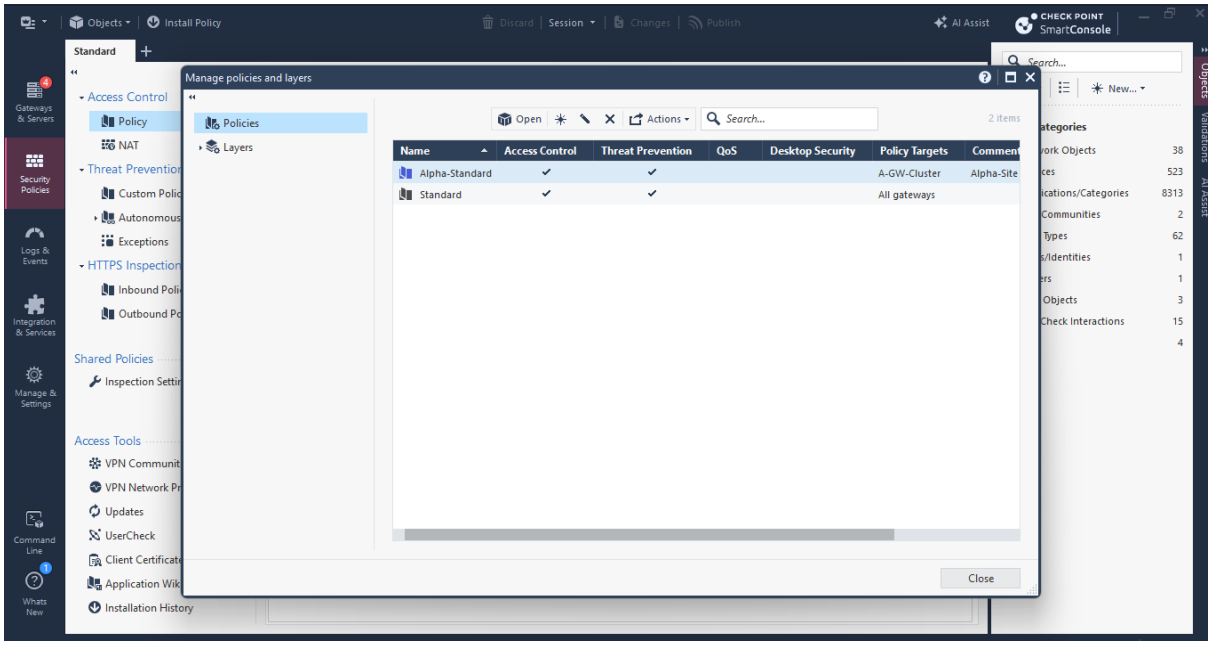
Brief Explanation: The **Hits**, **Track** and **Install On** columns are shown by checking the check box



Brief Explanation: The Source column is hidden and shown again

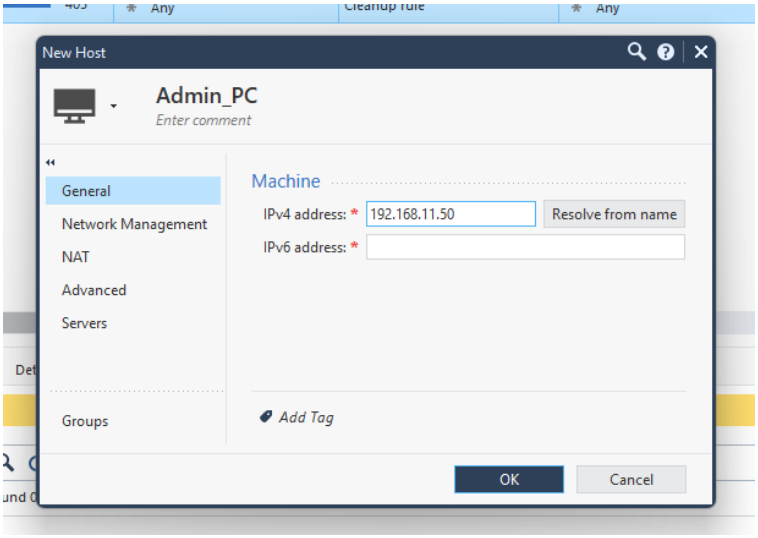


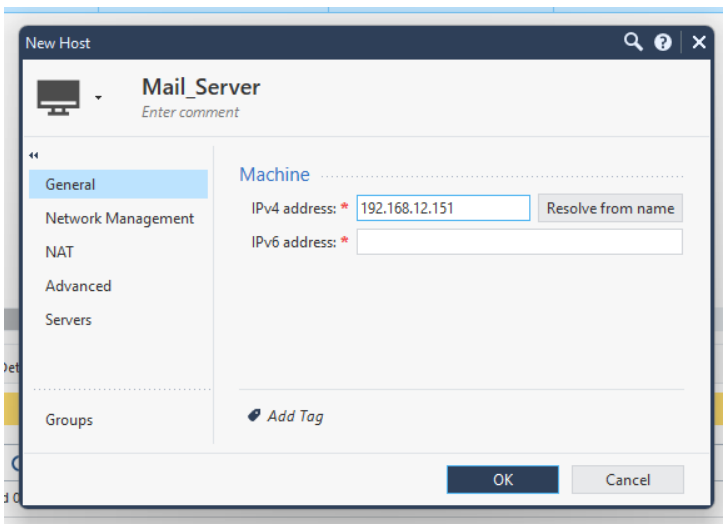
Brief Explanation: The three categories of setting are **Firewall, VPN,** and **Remote Acess**



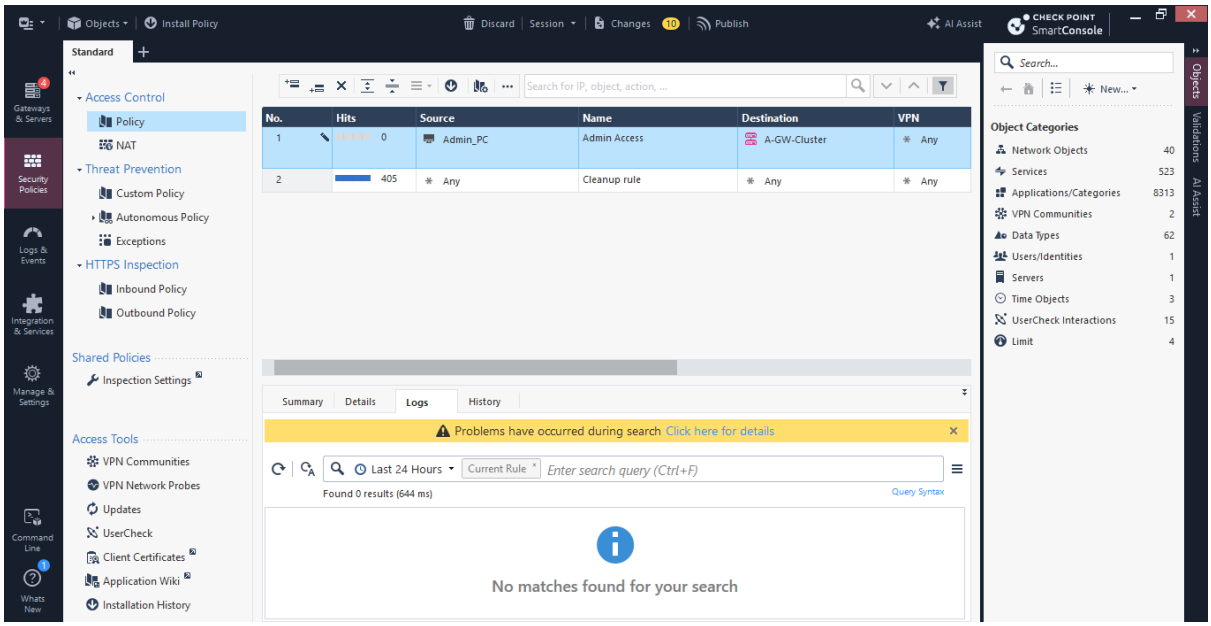
Brief Explanation: The Police package that exit include **Alpha-Standard** and **Standard**

TASK 2: BUILD THE ACCESS CONTROL RULEBASE



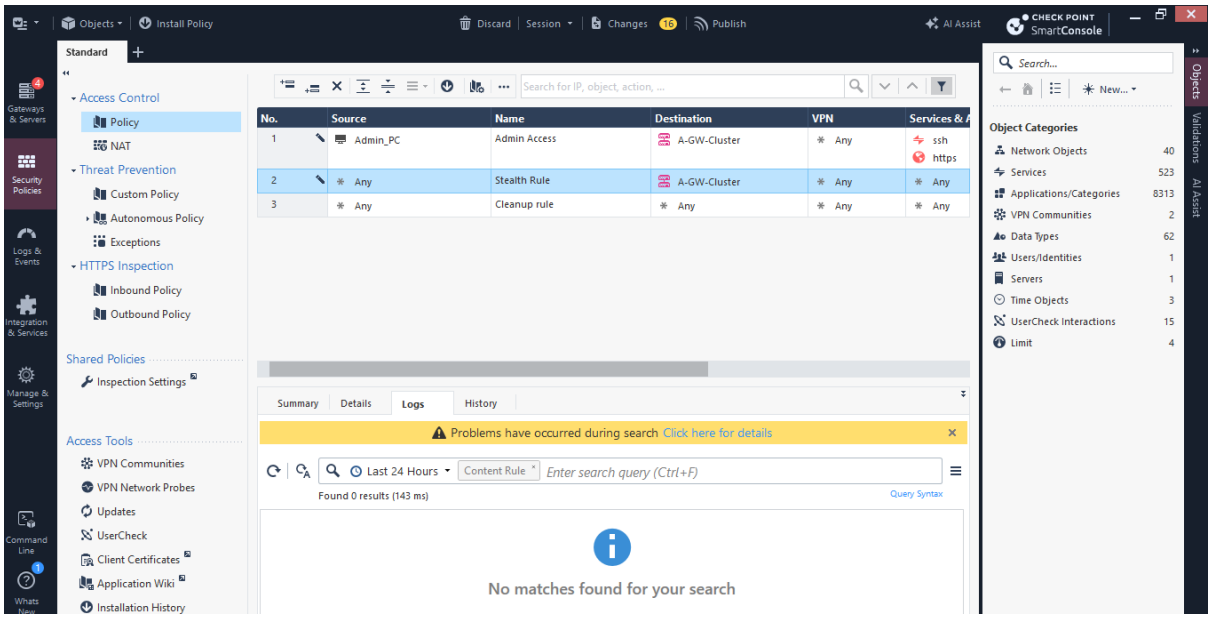


Brief Explanation: The Lab required Gateway, Web_Server, Internal_LAN, External_DNS, Admin_PC and Mail_Server but the **Admin_PC** and **Mail_Server** object were not found that is why we create them with the attached ip addresses



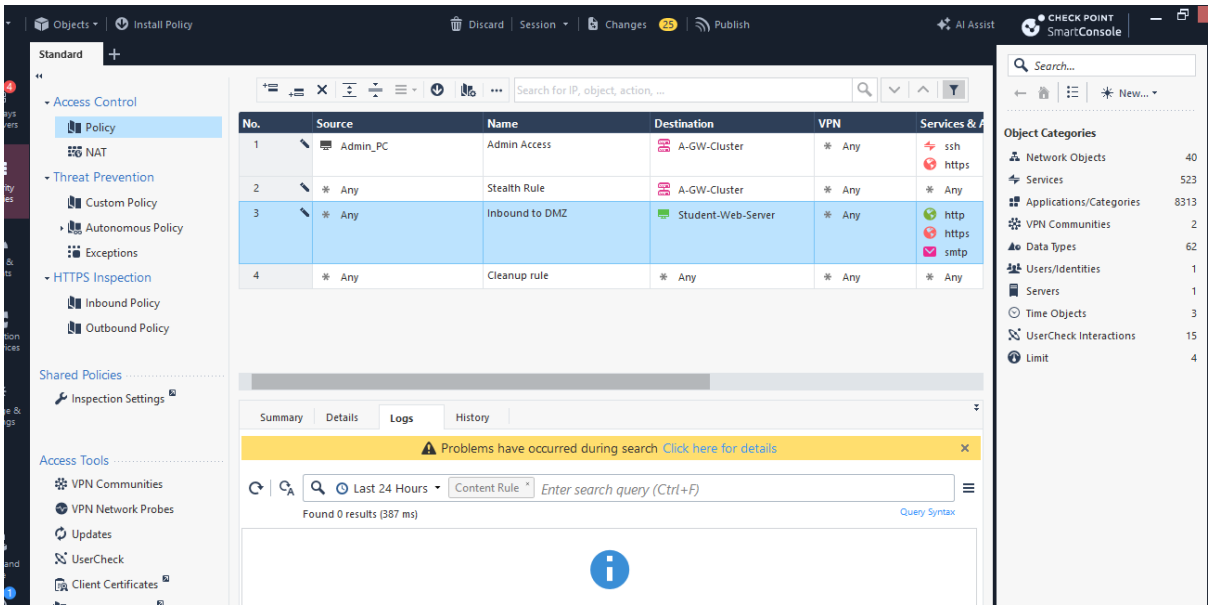
Brief Explanation: New rule is added above the cleanup rule with the following configuration: Name = Admin Access, Source = Admin PC, Destination = A-GW-Cluster, Services & Applications = ssh and https, Action = Accept, Track = Log

This rule allows the administrator which is **Admin_PC** to remotely manage the firewall (**A-GW-Cluster**) through ssh and https and the rule is place above cleanup rule so that administrator access will never be block accidentally



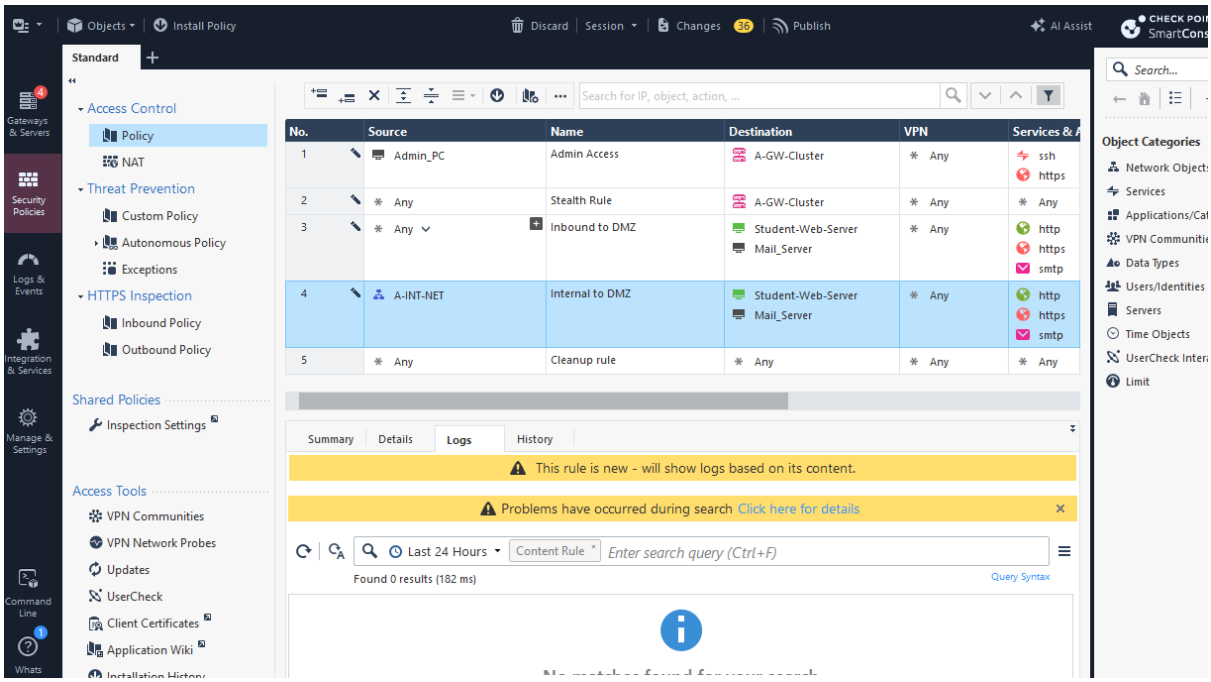
Brief Explanation: New rule is added above the cleanup rule with the following configuration: Name = Stealth Rule, Source = Any, Destination = A-GW-Cluster, Services & Applications = Any, Action = Drop, Track = Log

The **Stealth Rule** prevent any other user from accessing the firewall except the admin PC which is permit by a previous rule



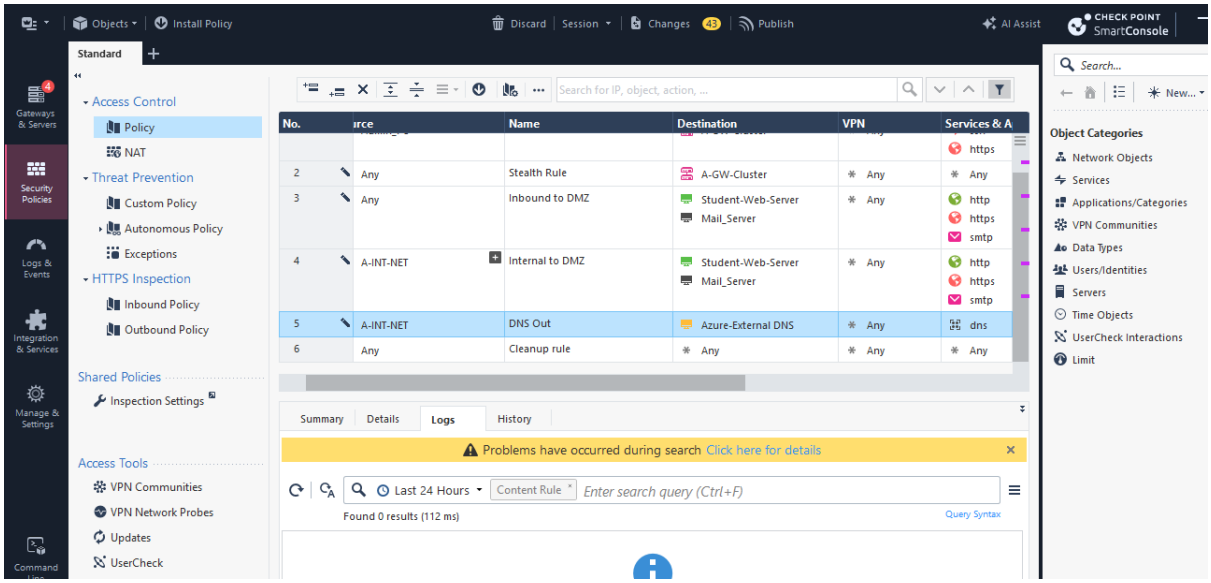
Brief Explanation: New rule is added above the cleanup rule with the following configuration: Name = Inbound to DMZ, Source = Any, Destination = Student-Web-Server, Mail_Server, Services & Applications = http, https, smtp, Action = Accept, Track = Log

This rule allows users from outside networks to access services hosted in the **DMZ**



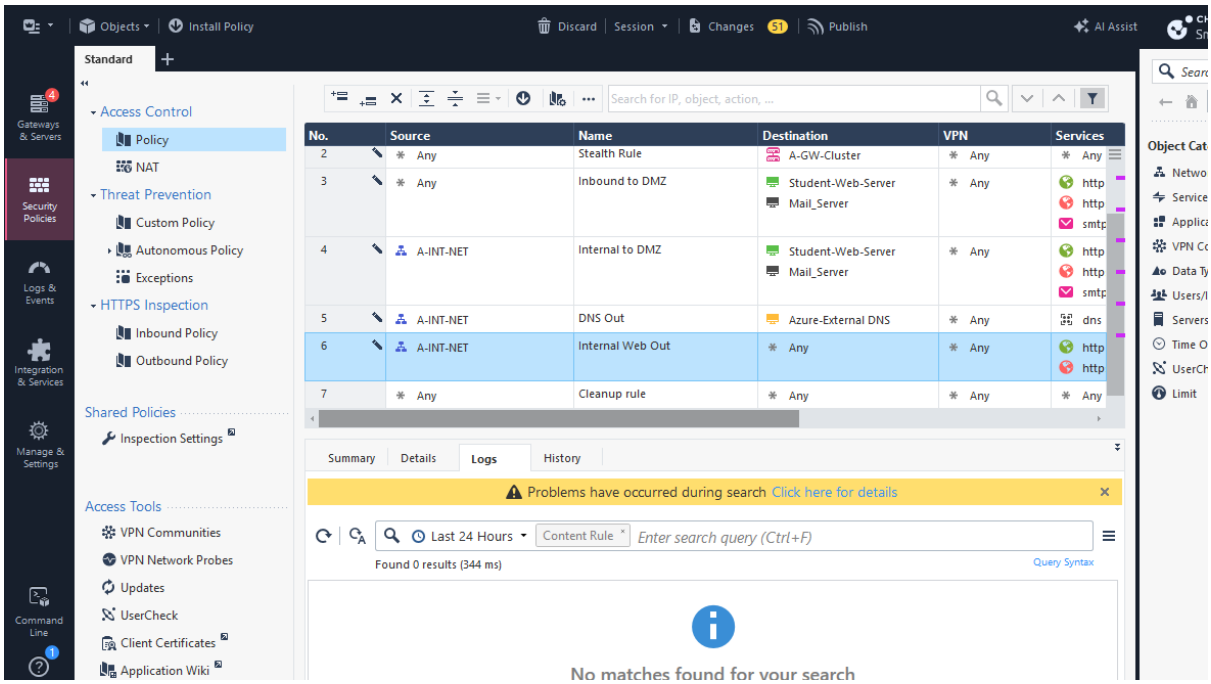
Brief Explanation: New rule is added above the cleanup rule with the following configuration: Name = Internal to DMZ, Source = A-INT-NET, Destination= Student-Web-Server, Mail_Server, Services & Applications = http, https, smtp, Action = Accept, Track = Log

This rule allows users on the **internal network** to access services hosted on the **DMZ servers**



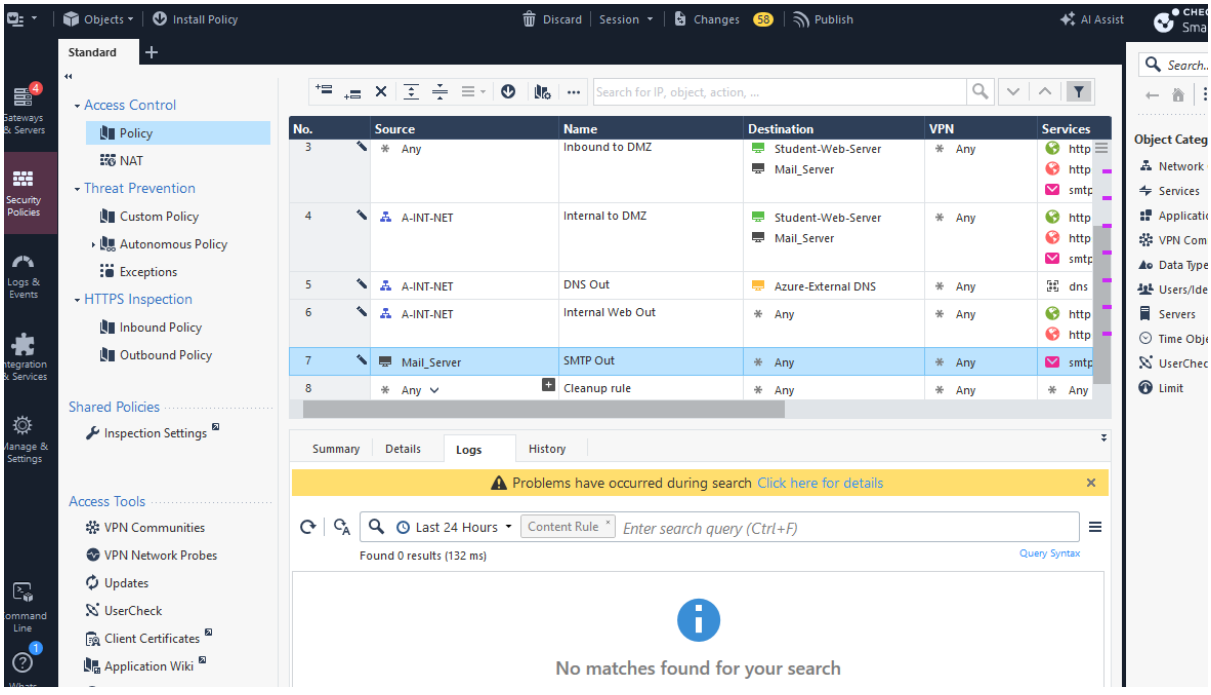
Brief Explanation: New rule is added above the cleanup rule with the following configuration: Name = DNS Out, Source = A-INT-NET, Destination = Azure-External DNS, Services & Applications = dns, Action = Accept, Track = Log

The **DNS Out** rule allows devices on the internal network to perform **DNS lookups** using the **Azure-External DNS** server



Brief Explanation: New rule is added above the cleanup rule with the following configuration: Name = Internal Web Out, Source = A-INT-NET, Destination = Any, Services & Applications = http and https, Action = Accept, Track = Log

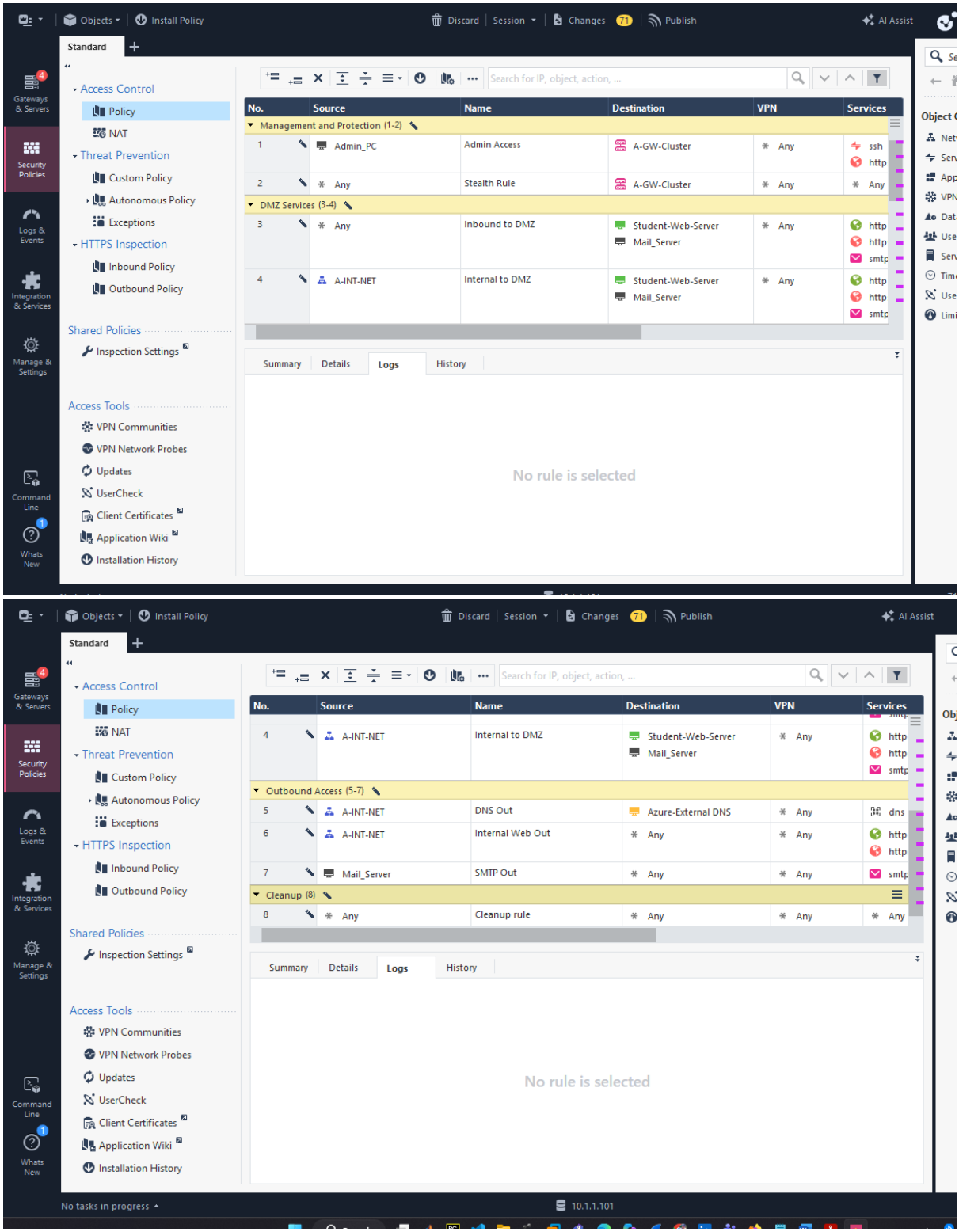
This rule allows users on the **internal network** to browse websites on the Internet.



Brief Explanation: New rule is added above the cleanup rule with the following configuration: Name = SMTP Out, Source = Mail_Server, Destination = Any, Services & Applications = smtp, Action = Accept, Track = Log

This rule allows your **Mail_Server** to send outgoing email to other mail servers on the Internet.

TASK 4: ORGANISE THE RULEBASE WITH SECTIONS

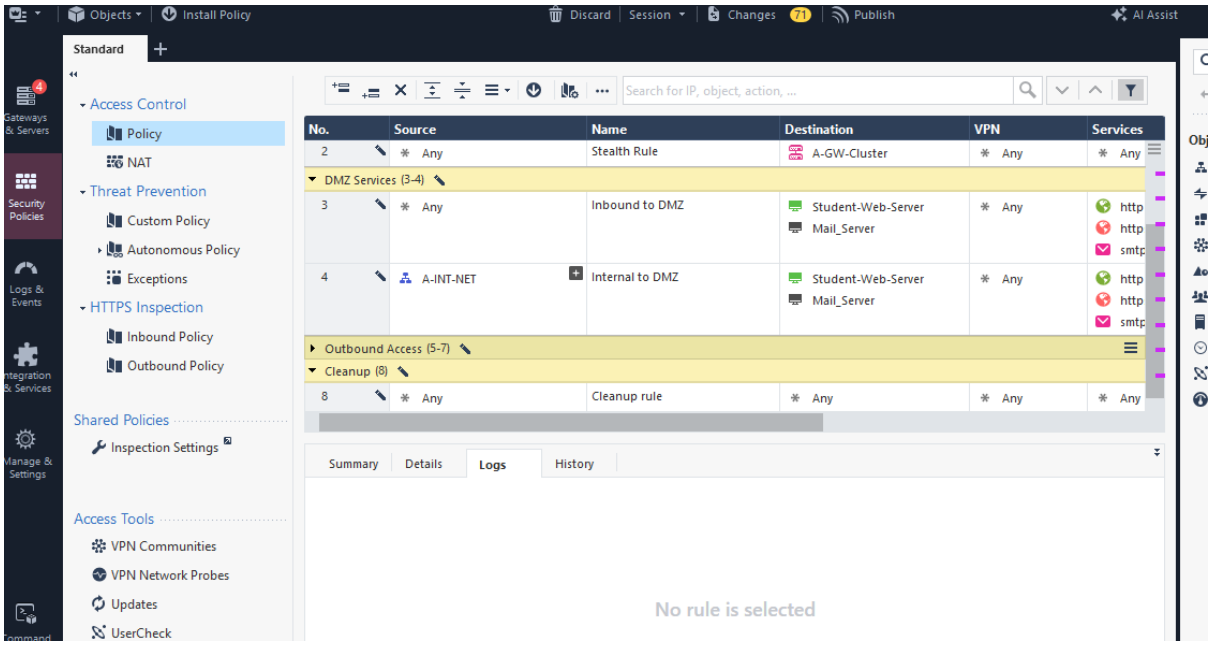


The Management and Protection section groups rules that protect and manage the firewall itself. Admin Access allows authorized administration while Stealth Rule blocks unauthorized access to the gateway.

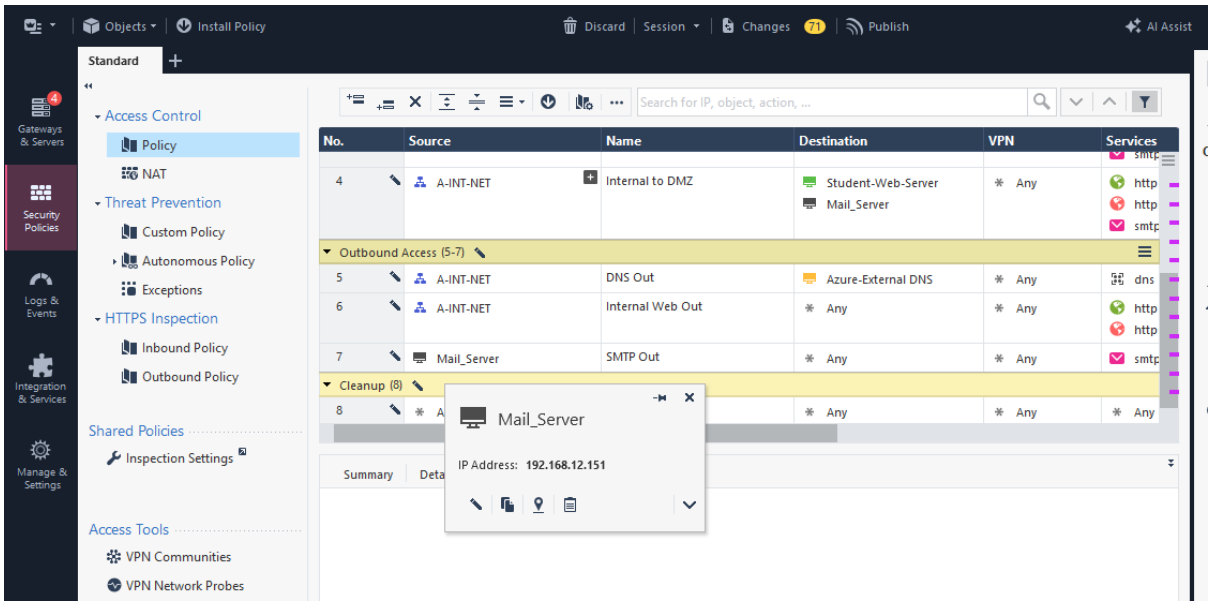
The DMZ Services section controls access to servers located in the DMZ. Public users access web and mail services while Internal users access web and mail services.

The Outbound Access section controls traffic leaving the organization. DNS queries, Web browsing and Outbound email

The Cleanup section is the final security section that blocks any traffic that is not permitted



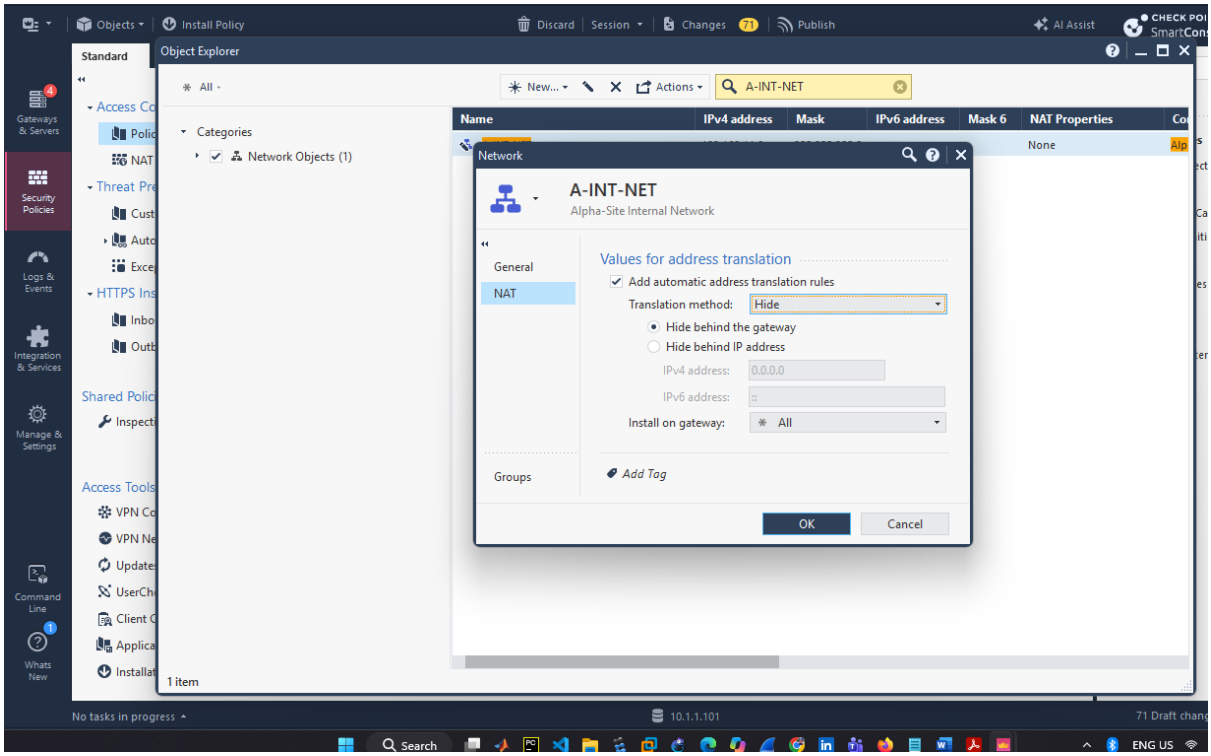
Outbound Access when collapsed



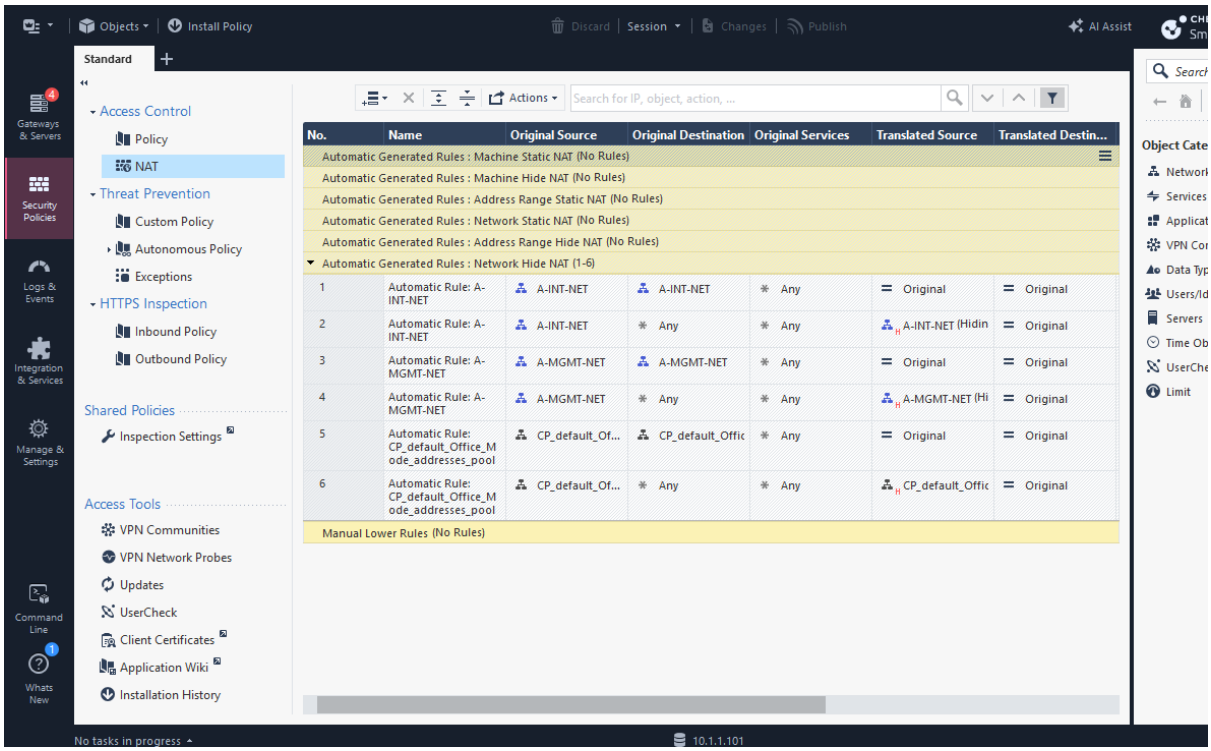
Outbound Access when expand

Finally, Sections are used to organize and manage the firewall rule base. They do not affect rule enforcement because the Security Gateway still evaluates rules from top to bottom, they are administrative labels and are not install on a security gateway

TASK 6: CONFIGURE HIDE NAT

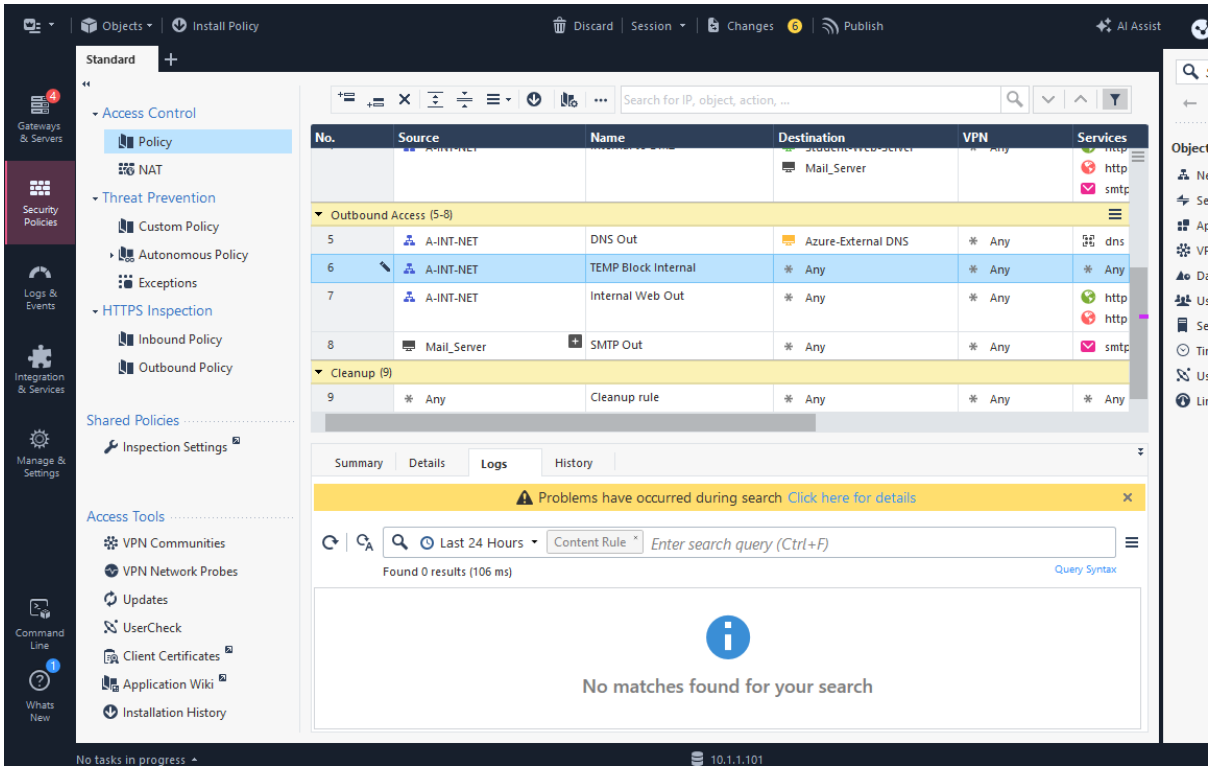


Brief Explanation: The **Hide behind Gateway** enable all hosts inside 192.168.11.0/24 to appear on the Internet using the gateway's public IP address



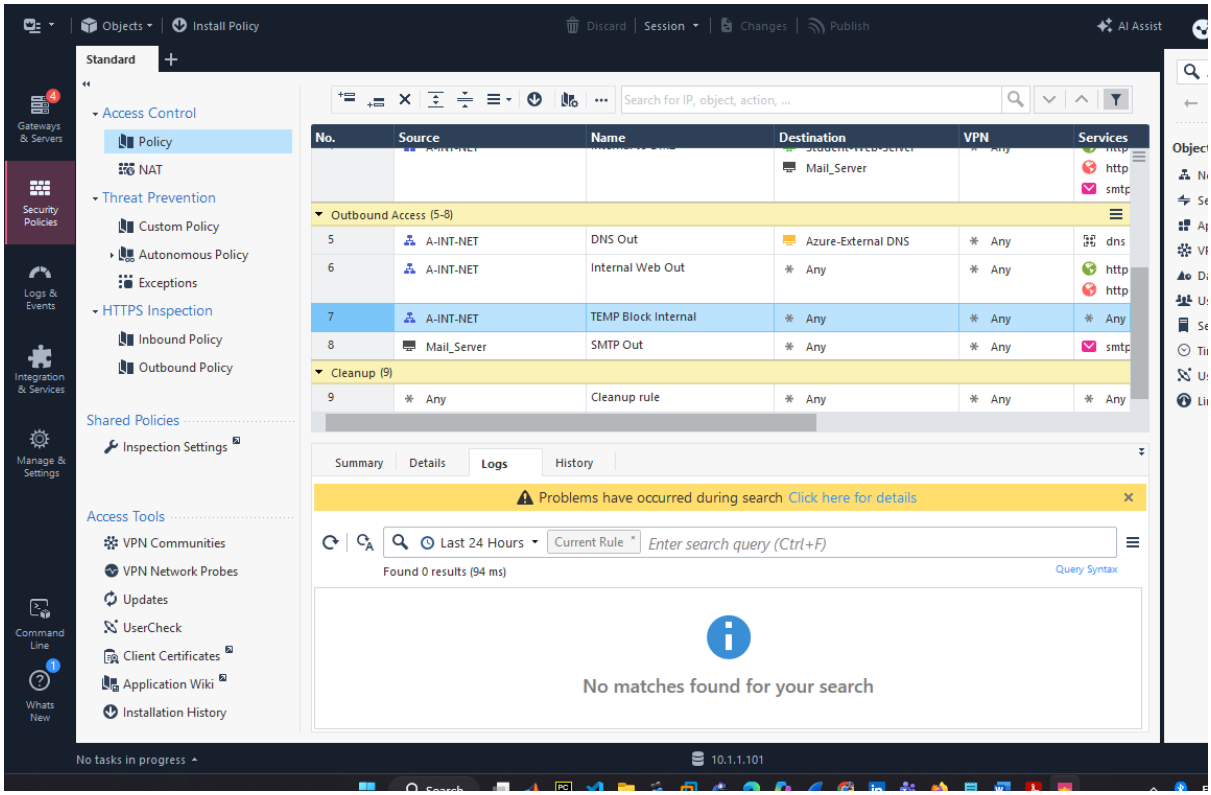
Brief Explanation: The NAT rule was created automatically when Hide NAT was enabled on the A-INT-NET network object, so there was no need to manually configure it in the NAT policy. Hide NAT uses a many-to-one translation method, which allows multiple devices on the internal network to share the gateway's public IP address when accessing external networks.

TASK 7: RULE ORDER EXPERIMENT AND INSTALLATION HISTORY



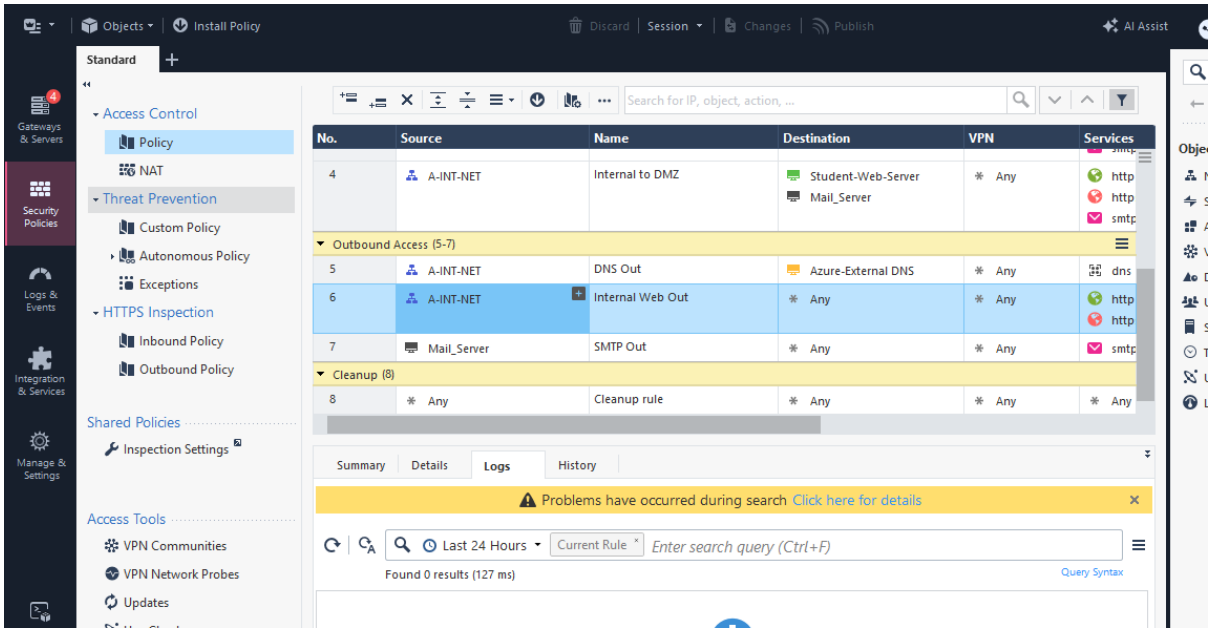
Brief Explanation: New rule is added above the Internal Web Out rule with the following configuration: Name = TEMP Block Internal, Source = A-INT-NET, Destination = Any, Services & Applications = Any, Action = Drop, Track = Log

This rule tells the firewall to Drop all traffic from A-INT-NET to anywhere and because it is **above Internal Web Out**, the firewall will hit this rule first and stop processing further rules

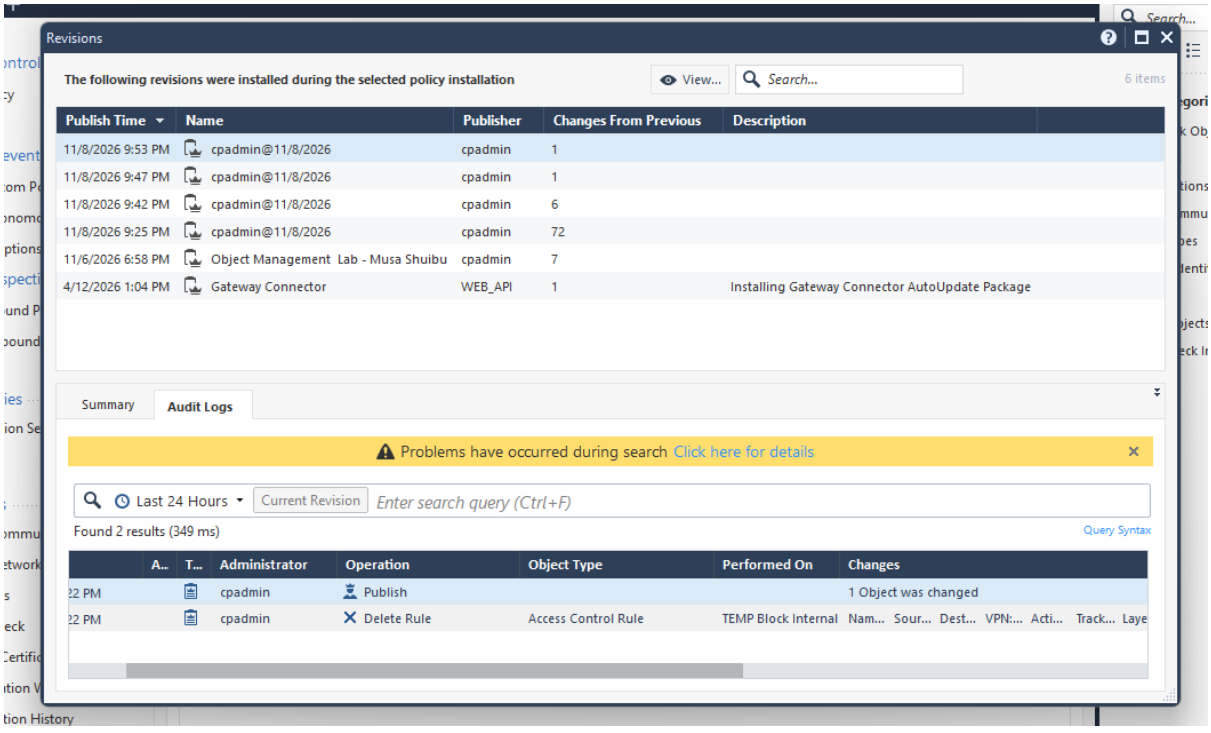


Brief Explanation: Now the web traffic will match rule 6: Internal Web Out before it reaches the drop rule now firewall stops processing rules as soon as it finds a match when TEMP Block Internal is above Internal Web Out, traffic is dropped immediately but now that the

TEMP Block Internal is moved below Internal Web Out, the traffic matches the allow rule first and is permitted



Now the temporary rule is deleted



SECTION B1:

Q1. Where is a Security Policy created, stored and enforced?

A. Created on the Gateway, stored in SmartConsole, enforced by the Management Server

B. Created in SmartConsole, stored on the Security Management Server, enforced by Security Gateways

C. Created on the Gateway, stored on the Gateway, enforced by SmartConsole

D. Created on the Management Server, stored on the Gateway, enforced by SmartConsole

ANS: B

Q2. Which control model does the Check Point Access Control firewall policy use?

A. Positive Control Model (whitelist)
B. Negative Control Model (blacklist)
C. A random mix of both models
D. Neither, because it only logs traffic

ANS: A

Q3. Where should the Stealth rule be placed?

A. At the very bottom of the rulebase
B. Above all rules, including Management rules
C. As early as possible, typically immediately after the Management rules
D. Directly above the Cleanup rule

ANS: C

Q4. Which statement about rulebase sections is correct?

A. Sections change the order in which rules are enforced
B. Sections are sent to the Security Gateway with the policy
C. Sections are required before zones can be used
D. Sections are visual divisions only and do not affect enforcement order

ANS: D

Q5. What type of translation does Hide NAT perform?

A. One-to-one
B. Many-to-one
C. One-to-many
D. Many-to-many with no ports involved

ANS: B

SECTION B2: SHORT ANSWERS

Q6. Distinguish between Explicit rules and Implied rules. State where Implied rules are configured and where they can be placed. What is the Implicit Cleanup Action?

ANS: Explicit rules are rules created manually by an administrator, while **implied rules** are automatically provided by Check Point for essential system functions. Implied rules are configured in **Global Properties** and can be placed before or after the rule base. The **Implicit Cleanup Action** is **Drop**, which means any traffic that does not match a rule is automatically blocked.

Q7. Explain why rule order matters in a rulebase. Give one example from your own rulebase.

ANS: Rule order is important because the firewall checks rules from top to bottom and applies the first rule that matches the traffic. If a more general rule is placed above a specific rule, it may match the traffic first and prevent the specific rule from being used. For example, when **TEMP Block Internal** was placed above **Internal Web Out** in the rulebase, web traffic from **A-INT-NET** was blocked; after moving it below **Internal Web Out**, the same traffic was allowed because it matched the web access rule first.

Q8. Explain the difference between saving and publishing changes in SmartConsole. What happens to unpublished changes when you install a policy, and why do many organisations publish only during a change window?

ANS: Saving changes in SmartConsole stores them in your current session, while **publishing** makes those changes available to other administrators and ready for policy installation. Unpublished changes are not included when a policy is installed, so only published revisions are enforced on the Security Gateway. Many organizations publish changes only during approved change windows to reduce the risk of disruption, allow proper testing and review, and maintain better control over production environments.

Q9. Compare Global Properties with Policy Packages. Give one example of what each controls and say how you open each in SmartConsole.

ANS: Global Properties contain settings that apply across the entire Check Point environment, while **Policy Packages** contain the security rules that control network traffic. For example, Global Properties can manage system-wide while Policy Package controls rules like **Admin Access**.

In SmartConsole, Global Properties can be opened through **Manage & Settings → Global Properties**, while a Policy Package can be accessed from **Security Policies**

Q10. Name the four predefined Security Zones. State two rules about assigning gateway interfaces to zones, and one benefit of using zone-based rules.

ANS: The four predefined Security Zones are **Internal**, **External**, **DMZ**, and **Wireless**. A gateway interface can belong to only one security zone, and all interfaces assigned to the same zone should have the same trust level. One benefit of using zone-based rules is that they simplify policy management by allowing administrators to create rules based on zones instead of individual networks or hosts.

Q11. A new administrator inserts a rule with Source Any, Destination Any, Services Any and Action Accept **above** the Stealth rule. Describe two security problems this causes, explain how to correct it, and name the type of rule that is allowed to sit above the Stealth rule and why.

ANS: Placing an **Any-Any-Any Accept** rule above the **Stealth Rule** creates two major security problems. First, it allows all traffic to pass through the firewall, which defeats the purpose of the security policy. Second, it allows unauthorized users to access the gateway, bypassing the protection provided by the Stealth Rule. To fix this issue, the overly permissive rule should be removed or moved below the Stealth Rule so that gateway protection is enforced first. The type of rule that is allowed above the Stealth Rule is a **management rule**, such as **Admin Access**, because administrators need secure access to manage the gateway using services like SSH and HTTPS.

SECTION B3: SCENARIO QUESTIONS

Q12. After Hide NAT is configured for Internal_LAN, staff report they cannot reach the Internet.

(a) State two benefits of NAT.

ANS:

NAT conserves public IP addresses by allowing multiple internal devices to share a single public IP address.

NAT improves security by hiding internal IP addresses from external networks, making the internal network structure less visible.

(b) Describe three checks you would make to find the fault, and explain why each one matters.

ANS:

1. Check that **Hide NAT is enabled** on the **A-INT-NET** object and configured to hide behind the gateway. If NAT is not configured correctly, internal hosts will not be translated when accessing the Internet.

2. Check that the **automatic NAT rule** exists in the NAT Rulebase. If the rule is missing or incorrect, the gateway will not perform address translation for internal users.

3. Check that the required **Access Control rules** are present and installed, especially **DNS Out** and **Internal Web Out**. Even if NAT is working, users will not be able to reach Internet resources if the firewall policy blocks DNS or web traffic.